# Climate Shocks, Health Finance, and Economic Outcomes

County-Level Evidence from the United States, 2011–2023

Seminar Presentation · April 2026

# Climate Science Has Cracked Mortality. Health Finance Is the Blind Spot.

**The economic burden of climate on healthcare costs and household debt is largely unmeasured**

## What we know well

- Extreme heat, cold, and drought raise **mortality** — robust evidence across many countries
- Air pollution ($PM_{2.5}$, ozone) shortens lives and raises hospitalisation rates
- Climate models project these risks will intensify

### What we don't know

- Does surviving a climate shock leave households **financially damaged**?
- Do insurers and hospitals **absorb or pass on** those costs?
- Does climate volatility — not just severity — matter for **health finance**?

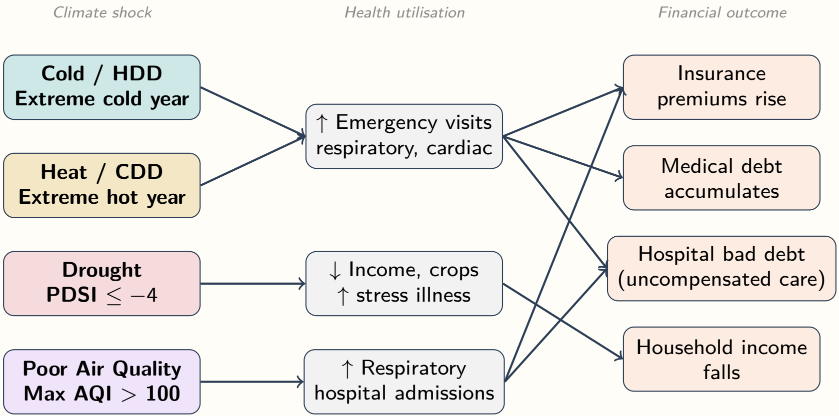
## Why this matters for adaptation

- Adaptation policy that targets mortality alone may **miss the majority of the economic burden**
- Medical debt and insurance premiums are **regressive**: low-income households bear disproportionate shares
- Hospital bad debt signals coverage collapse under climate stress — an early warning for **system fragility**
- Evidence is needed at the **county scale** where adaptation decisions are made

This paper: first systematic county-level evidence linking climate shocks to health finance outcomes across the U.S., 2011–2023.

# How Climate Shocks Reach Your Wallet

## The transmission chain from weather event to household financial damage




This paper measures the **right-hand column** directly using county panel data, bypassing utilisation data which are noisy and inconsistently reported.

# What Are We Measuring? A Field Guide to Health Finance Outcomes

## Plain-English definitions; all dollar values in $2023

| Outcome | What it measures | Scale / context |
|---|---|---|
| **Benchmark Silver Premium** | Monthly cost of the mid-tier ACA marketplace plan in a county's rating area. Insurers set this annually based on expected claims. | ∼$400–700/month nationally; +$28 = ∼4–7% increase |
| **Medical Debt Share** | Fraction of county residents with medical debt in collections, from credit bureau records (Urban Institute). | National mean ∼13%; +1 pp = ∼8% relative rise |
| **Medical Debt Median** | Median balance owed among those with medical debt in collections. | National median ∼$690; +$30 ≈ 4% increase |
| **Hospital Bad Debt per Capita** | Uncompensated care written off by hospitals after billing failure (not charity care). Signals patients unable to pay even after treatment. | Mean ∼$120/capita; +$5 ≈ 4% increase |
| **Medicaid / Medicare per Enrollee** | Annual public payer spending per enrolled beneficiary. Falls can indicate access barriers, not just lower need. | Medicare ∼$12,000/yr; Medicaid ∼$8,000/yr |
| **Median HH Income / PCPI** | County median household income (ACS) and BEA per-capita personal income. Both inflation-adjusted. | U.S. median HH income ∼$75,000 in 2023 |

# Data: Linking Climate Records to Health Finance at County Level

**Panel: 2011–2023; ∼3,155 U.S. counties; state panel: 51 states × 13 years**

## Climate & Environment

| Variable | Source |
|---|---|
| Annual mean temp, HDD, CDD | NOAA/NCEI |
| Palmer Drought Severity Index | NOAA/NCEI |
| Annual AQI (median, max) | EPA |

### Health Finance

| Variable | Source |
|---|---|
| ACA marketplace premiums | HIX Compare |
| County medical debt | Urban Institute |
| Hospital bad / charity debt | NASHP HCT |
| Employer insurance contrib. | AHRQ MEPS-IC |
| Medicaid / Medicare spending | CMS NHE |

## Economy & Household

| Variable | Source |
|---|---|
| Median HH income | ACS 5-yr |
| Civilian employment | ACS 5-yr |
| Per-capita personal income | BEA CAINC1 |
| CPI (inflation adjustment) | FRED |

### Key features

- Climate z-scores anchored to **1990–2000 baseline** — shocks measured against pre-warming reference
- CDD/HDD thresholds: **national 80th percentile** — captures objectively extreme years, not just "warm for that county"
- All dollar outcomes **inflation-adjusted to 2023**
- State-level cluster SEs account for **within-state spatial correlation**

# Empirical Specification

**Two-way fixed effects with distributed lags; all dollar outcomes inflation-adjusted to $2023**

## Estimating Equation

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^{2} \beta_h \,\mathrm{Shock}_{i,t-h} + \mathbf{X}'_{it}\delta + \varepsilon_{it}$$

- $\alpha_i$: unit FE (state or county); $\gamma_t$: year FE
- Distributed lags $h = 0, 1, 2$ for all shocks
- SE clustered at **state level** throughout
- **County:** also rating-area-clustered SEs for premiums

## Climate Shock Variables

| Variable | Definition |
|---|---|
| `High_CDD` | CDD $\geq$ national p80 (1990–2000) |
| `High_HDD` | HDD $\geq$ national p80 (1990–2000) |
| `Z_Temp` | $(T_{it} - \bar{T}_i^{90\text{-}00})/\sigma_i^{90\text{-}00}$ |
| `Z_Precip` | Precipitation z-score (same baseline) |
| PDSI | Palmer drought severity index |
| `High_AQI_Max` | Max AQI $>$ 100 (binary) |

## Outcome Blocks

| Outcome | Source |
|---|---|
| Employer premium contribution | MEPS-IC |
| Benchmark Silver premium | HIX |
| Medical debt share / median | Urban Inst. |
| Hospital bad debt per capita | NASHP |
| Total, Medicaid, Medicare spending | CMS |
| Median HH income / employment | ACS |
| Per-capita personal income | BEA |

# Dynamic Estimation Strategy

**Baseline dynamic-panel distributed lags and local projections recover complementary timing margins**

## Dynamic Panel / Distributed Lag (DP/DL)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^{2} \beta_h \,\mathsf{Shock}_{i,t-h} + \mathbf{X}_{it}'\delta + \varepsilon_{it}$$

- Main FE estimator for contemporaneous and lagged effects
- Cumulative response is the sum of lag coefficients
- Baseline estimator for the state and county tables

## Local Projections (LP, Jordà 2005)

$$Y_{i,t+h} = \alpha_i^h + \gamma_t^h + \beta_h \,\mathsf{Shock}_{i,t} + \mathbf{X}_{it}'\delta_h + \varepsilon_{i,t+h}$$

- Separate regression for each horizon $h \in \{-2, \ldots, +3\}$
- Negative horizons serve as placebo leads for pre-trend checks
- More flexible when lag shapes are misspecified

**Controls in both approaches:** uninsured rate, median household income, and the same county/year fixed effects.
**Inference:** state-clustered SEs throughout; rating-area clustered variants for premium outcomes as robustness.

# Identification: What Does Two-Way Fixed Effects Buy Us?

The key assumption and what variation we exploit

## The Problem

Counties with more extreme climate also differ in many other ways: poorer infrastructure, older populations, fewer hospitals. A naïve comparison of high-climate-shock vs. low-climate-shock counties would pick up all of these differences.

### The Fix: Two-Way Fixed Effects

- **County FE** ($\alpha_i$): removes all time-invariant county characteristics. We only use *within-county* variation across years.
- **Year FE** ($\gamma_t$): removes common nationwide trends (recessions, ACA roll-out, COVID). We only use *deviations from national trends*.
- **Net result:** we compare a county to *itself* in more vs. less extreme climate years, after removing the national year trend.

## Intuition with an Example

Phoenix, AZ always has extreme heat. A county FE removes Phoenix's perpetually high premiums. We ask: **were Phoenix premiums unusually high in years when CDD was especially elevated relative to Phoenix's own baseline?**

### Distributed Lags

- Health costs don't always respond immediately
- Insurers set premiums *before* a plan year (anticipatory)
- Medical debt accumulates *after* illness (delayed)
- We include $h = 0, 1, 2$ to capture the full dynamic response

# Robustness: Pre-Trend Checks and Multicollinearity Diagnostics

Confirming that results are not driven by pre-existing trends or collinear regressors

## Local Projection Pre-Trend Test

- LP at **negative horizons** ($h = -2, -1$) tests whether outcomes were already trending *before* the shock
- Under the parallel-trends assumption, pre-shock coefficients should be near zero and insignificant
- In our data: **pre-trend coefficients are consistently small and not significant** for key relationships (HDD $\rightarrow$ premiums, drought $\rightarrow$ PCPI)
- Formally: Wald test of joint pre-trend significance fails to reject the null

## Multicollinearity (VIF)

- PDSI, PHDI, and PMDI are highly correlated drought indices
- Including all three inflates Variance Inflation Factors
- **Primary specs use PDSI only**; PHDI/PMDI retained for optional robustness
- Post-pruning: max VIF $\approx 5.3$ across all county specs — well within acceptable range

### Premium Clustering

- Benchmark Silver premiums are set at the **rating-area** level — counties in the same rating area have *identical* premiums by construction
- Primary SEs are state-clustered (conservative)
- Rating-area-clustered variants reported for all premium outcomes as robustness — results hold

# Roadmap: Three Complementary Analyses

**This presentation covers state, county, and a new weather-swing analysis**

## State-Level FE

- 51 states $\times$ 13 years
- Drought and cold channels on premiums and debt
- Establishes the aggregate picture

## County-Level FE

- $\sim$3,155 counties, 2011–2023
- HDD, CDD, drought, AQI on debt, premiums, income, hospital costs
- Higher power; rating-area clustering for premiums

## Weather Swing (New)

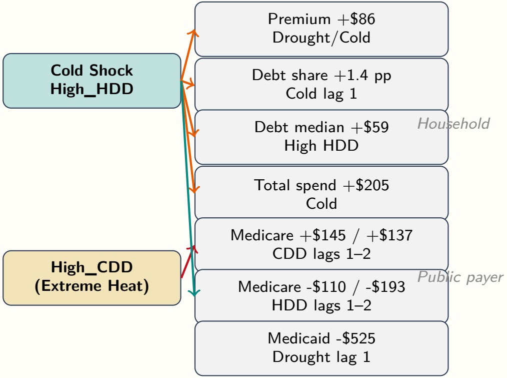
- County-level only
- Exposure: $\Delta X_{it} = X_{it} - X_{i,t-1}$
- Identifies cost of **volatility**, not just level
- Finds effects where level specs were insignificant (AQI)

State panel ($N \approx 650$) has insufficient power for first-difference specifications — swing analysis is county-only.

# State-Level Results

Two-way FE: State + Year, $N \approx 550$–$650$, cluster = State

# State-Level: Cold and Heat Drive Opposite Financial Channels




**Cold channel:**

- Premiums, debt, and total spending rise
- Medicare falls with lagged HDD exposure

**Heat channel (CDD):**

- Medicare rises 1–2 years later

**Drought:**

- Medicaid falls and premiums rise

Heat and cold affect public payers in opposite directions — novel finding.

# State: Drought and Cold Raise Premiums and Debt

**Household & individual outcomes;** $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$

## Employer Premium Contribution ($/yr)

| Predictor | Coef. | |
|---|---|---|
| Extreme Drought ($h$=0) | +63.6 | ** |
| Severe Drought ($h$=0) | +86.4 | *** |
| Severe Drought (lag 1) | +65.1 | ** |
| Pct $PM_{2.5}$ days | +20.8 | *** |
| Pct $PM_{10}$ days | +22.7 | *** |
| Pct Ozone days | +19.7 | *** |

Air quality pollutant days show consistent positive effects on employer premium costs.

## Medical Debt Share (pp)

| Predictor | Coef. | |
|---|---|---|
| Cold Shock (lag 1) | +0.0135 | ** |

## Medical Debt Median ($/yr)

| Predictor | Coef. | |
|---|---|---|
| High HDD ($h$=0) | +58.9 | ** |
| Severe Drought ($h$=0) | −66.8 | *** |
| Severe Drought (lag 1) | −37.5 | ** |
| Heat Shock ($h$=0) | +32.7 | * |
| Heat Shock (lag 2) | +47.7 | ** |

Cold raises debt prevalence; drought paradoxically reduces median severity.

# State: Cold Raises System Costs; Drought Cuts Public Payer Spending

Hospital & system outcomes; two-way FE (State + Year)

### Total Per-Capita Health Expenditure ($)

| Predictor | Coef. | |
|---|---|---|
| Cold Shock ($h$=0) | +205.3 | ** |
| High HDD (lag 2) | +92.0 | * |

### Medicaid per Enrollee ($)

| Predictor | Coef. | |
|---|---|---|
| Extreme Drought ($h$=0) | −566.2 | ** |
| Severe Drought ($h$=0) | −525.8 | ** |
| Severe Drought (lag 1) | −524.5 | *** |
| Severe Drought (lag 2) | −288.4 | ** |

Drought

persistently reduces Medicaid spending across lags 0–2.

### Medicare per Enrollee ($)

| Predictor | Coef. | |
|---|---|---|
| Severe Drought (lag 1) | −144.3 | *** |
| High CDD (lag 1) | +144.7 | ** |
| High CDD (lag 2) | +137.2 | *** |
| High HDD (lag 1) | −110.1 | * |
| High HDD (lag 2) | −192.7 | *** |
| Unhealthy days | +97.7 | ** |

Heat

(CDD) raises Medicare with lag; cold (HDD) reduces it — opposite channels.

# State Results: Plain-English Interpretation

What do these coefficients mean for a real state?

## Cold shock (High HDD) raises premiums

A year with extreme heating degree days raises the Benchmark Silver premium by **+$86/month** for severe drought conditions or around +$28–$63 depending on drought severity. For a family of four, that is **+$1,000–$4,000 per year** in insurance costs during a bad cold year.

### Drought suppresses Medicaid spending

Extreme drought reduces Medicaid per-enrollee spending by **–$525** in the same year and remains negative for two further years. This is *not* good news: it likely reflects **forgone care** — fewer doctor visits during economic stress, not lower medical need.

## Heat raises Medicare with delay

Extreme CDD years raise Medicare per-enrollee spending by **+$137–$145** at lags 1–2. The elderly are most vulnerable to heat-related illness; their costs appear in the Medicare budget 1–2 years after the shock (hospitalisation, follow-up care).

### Heat and cold push Medicare in opposite directions

Cold (HDD) *reduces* Medicare with lag (−$110 to −$193). The asymmetry suggests different mechanisms: cold stress triggers emergency use that is absorbed privately or via Medicaid, while heat stress generates Medicare-relevant chronic-disease exacerbation.

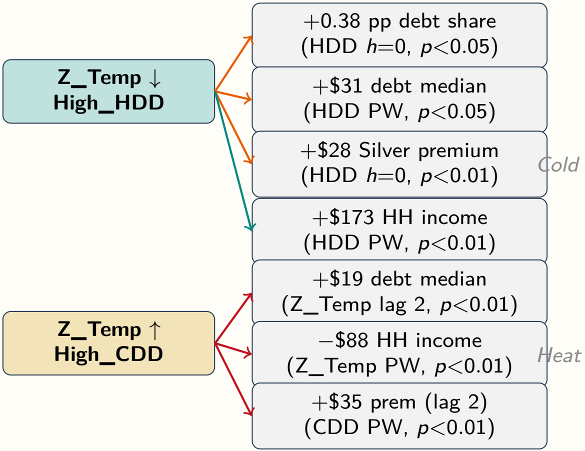
**Summary:** cold damages household and private insurance; drought damages the public safety net; heat shifts costs to Medicare.

# County Results: Household & Individual

Medical debt, premiums, income, employment

Two-way FE: County + Year, $N \approx 23k–34k$, cluster $=$ State

# County-Level: Cold and Heat Channels Mirror the State Findings




**Cold channel:**

- Raises debt prevalence and severity
- Raises premiums on impact
- Raises HH income (energy mix?)

**Heat channel:**

- Raises debt median (lagged)
- **Reduces** HH income contemporaneously
- Raises premiums with 1–2 yr lag

**Cross-level:**

- Cold → debt: both levels confirmed
- Heat → income: county only
- Premium channels align in direction

# County: Heat Raises Debt Severity; Cold Raises Debt Prevalence

Medical Debt Share (pp) and Median Debt (\$2023); UW = unweighted, PW = pop-weighted

## Z_Temp (Relative Heat Shock)

| | UW | PW |
|---|---|---|
| *Debt Share (pp)* | | |
| $h$=0 | −0.0025* | −0.0000 |
| Lag 2 | −0.0027** | −0.0014* |
| *Debt Median (\$)* | | |
| $h$=0 | +14.9* | +18.5*** |
| Lag 2 | +19.5** | +16.2*** |

Heat reduces prevalence but raises severity — consistent composition effect: warm years shed marginal debtors while pushing the remaining cases deeper.

## High_HDD (Extreme Cold Burden)

| | UW | PW |
|---|---|---|
| *Debt Share (pp)* | | |
| $h$=0 | +0.0038** | +0.0054** |
| Lag 2 | +0.0031* | +0.0037* |
| *Debt Median (\$)* | | |
| $h$=0 | +10.6 | +31.2** |

**Controls (all debt models):**

| | |
|---|---|
| Uninsured Rate | +*** **(large)** |
| HH Income | −** |

Cold raises both prevalence and severity — no composition effect.

# County: Cold Raises Premiums Now; Heat Raises Them Later

Benchmark Silver Premium ($2023 real); county + year FE

### High_HDD (Cold Burden)

| Lag | UW | PW | Cluster |
|---|---|---|---|
| $h$=0 | +28.1*** | +21.9** | State |
| $h$=0 | +28.1*** | +21.9*** | RA |
| Lag 1 | −24.5* | −20.8 | State |

### High_CDD (Heat Burden) — lagged

| Lag | UW | PW | Cluster |
|---|---|---|---|
| Lag 1 | +8.7 | +31.4*** | RA |
| Lag 2 | +21.6* | +35.3*** | RA |

### Z_Temp (Relative Temp)

| Lag | UW | PW | Cluster |
|---|---|---|---|
| $h$=0 | −10.9** | +2.7 | RA |
| Lag 1 | +3.8 | +5.4** | RA |
| Lag 2 | +1.0 | +4.2** | RA |

- **Cold (HDD)** raises premiums on impact in both state- and RA-clustered specs
- **Heat** lowers premiums now, then raises them 1–2 years later
- RA clustering tightens CIs; drought lag 1 is borderline at +$3.9 ($p$=0.05)

# County: Heat Depresses Income; Drought Hits PCPI

Median HH income and per-capita personal income ($2023)

## Median HH Income ($2023)

| Predictor | UW | PW |
|---|---|---|
| Z_Temp ($h$=0) | −71** | −88*** |
| Z_Temp (lag 1) | −36 | −64*** |
| Z_Temp (lag 2) | −71** | −60** |
| High_CDD ($h$=0) | −247** | −3 |
| High_HDD ($h$=0) | +161** | +173*** |

Relative

heat most consistently lowers household income in the population-weighted specifications.

## Per-Capita Personal Income ($)

| Predictor | UW | PW |
|---|---|---|
| PDSI lag 2 | −8.5 | −113** |
| High_CDD lag 1 | +480** | −116 |
| RA-level: PDSI ($h$=0) | −206*** | |

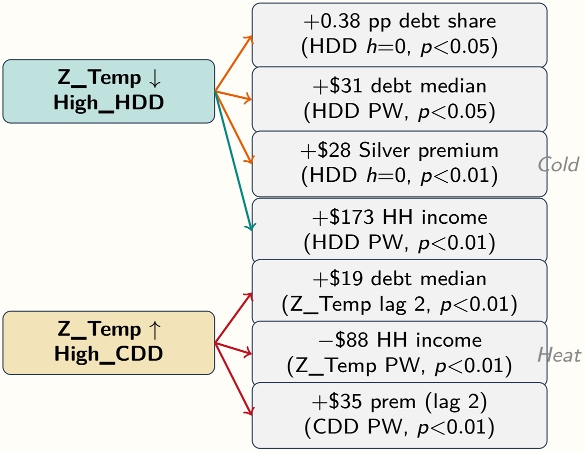
## Income Summary

- **Heat** lowers HH income across lags in pop-weighted models
- **Cold** raises HH income on impact, likely reflecting seasonal or energy-income composition
- **Drought** is strongest in PCPI: RA spec gives −$206***

## Civilian Employment (persons)

| Predictor | UW | PW |
|---|---|---|
| Z_Temp lag 2 | +788** | +3216 |
| High_HDD ($h$=0) | −468** | +2587 |
| High_HDD lag 2 | −678** | +3988 |
| Z_Precip ($h$=0) | +46 | −5742*** |
| Z_Precip (lag 1) | −88 | −3803*** |

# County-Level: Cold and Heat Channels Mirror the State Findings

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**Heat channel:**

- Raises debt median (lagged)
- **Reduces** HH income contemporaneously
- Raises premiums with 1–2 yr lag

**Cross-level:**

- Cold → debt: both levels confirmed
- Heat → income: county only
- Premium channels align in direction

# County vs. State: Why More Granularity Changes What We Find

**County FE absorbs within-state heterogeneity invisible to state models**

## What county models add

- State models average over 3,000+ counties — rural and urban, coastal and inland, insured and uninsured. County models let each county serve as its own control.
- **Heat → household income loss**: not visible in state models, emerges clearly in county models ($-\$88^{***}$ per capita). Likely masked by state-level averaging with opposite-direction rural/urban effects.
- **Cold → hospital bad debt**: county-level variation ($+\$5/capita$) is not detectable in the $N = 650$ state panel.
- **Rating-area structure**: premiums are set at the rating-area level. County models with rating-area clustering are the correct specification.

## Where state and county agree

- Cold → debt and premiums: confirmed at both levels
- Drought → public payer suppression: both levels significant
- Heat → delayed premium increases: both levels
- Air quality → employer premiums: state level (state AQI); county level uses Max AQI shock indicator

### Sample size and power

- State: $N \approx 650$; County: $N \approx 23{,}000$–$34{,}000$
- County panel has $\sim 50\times$ more observations
- Allows detection of effects too small to identify at state level

# County Results: Hospital Sector

Hospital bad debt per capita

Two-way FE: County + Year, $N \approx 27k$, cluster = State

# County: Cold Raises Hospital Bad Debt; Heat Lag Reduces It

**Hospital Bad Debt per Capita ($); within-$R^2 = 0.006$–$0.019$**

### Key Coefficient Estimates

| Predictor | UW | PW |
|---|---|---|
| Z_Temp lag 2 | $-2.41^{***}$ | $-2.08^{***}$ |
| Z_Temp ($h$=0) | $-1.25$ | $-1.08^{**}$ |
| High_HDD ($h$=0) | $+5.33^{***}$ | $+4.23^{**}$ |
| PDSI lag 1 | $+0.57^{*}$ | $+0.50^{*}$ |
| PDSI lag 2 | $+0.01$ | $+0.50^{*}$ |
| Uninsured Rate | $+151^{***}$ | $+142$ |

RA-level robustness (pop-wtd):

PDSI lag 2: $+0.76^{***}$ Z_Temp lag 2: $-2.26^{***}$

- **Cold (HDD) raises bad debt on impact** by $4–5 per capita — consistent with emergency utilisation, deferred billing, and coverage gaps during cold events
- **Heat lag reduces bad debt** (2-yr lag, $p<0.001$) — possible reduction in elective utilisation during hot years or survivor selection
- **PDSI (drought)** adds a smaller lagged increase, confirmed in RA robustness
- **Uninsured Rate** is the dominant control ($+151^{***}$ unweighted) — structural coverage gap drives hospital write-offs

About 23% of county-years lack hospital reports; one negative outlier is winsorized.

# Weather Swing Analysis

A new source of variation: year-over-year climate volatility

County + Year FE, $N \approx 23\text{k}–34\text{k}$, cluster = State

# New Analysis: Exploiting Year-to-Year Weather Swings

County-level only; $N \approx 650$ state-years insufficient for first-difference identification

## Motivation

- Prior specs identify counties in persistent shock states vs. not
- New margin: do counties that **swing sharply** year-to-year bear extra health costs, regardless of average climate?
- Is it **volatility**, not just level, that damages health finance?

### Robustness Specs

- **Asymmetric:** $\Delta^+ = \max(\Delta X, 0)$ vs. $\Delta^- = \min(\Delta X, 0)$
- **Binary transitions:** Onset / Exit / Persist indicators

## Primary Specification

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \underbrace{\Delta X_{it}}_{\text{swing}} + \beta_2 \underbrace{X_{i,t-1}}_{\text{lagged level}} + \mathbf{X}_{it}'\delta + \varepsilon_{it}$$

- $\Delta X_{it} = X_{it} - X_{i,t-1}$ for all continuous exposures: `Z_Temp`, `Z_Precip`, `CDD`, `HDD`, `PDSI`, `Median_AQI`, `Max_AQI`
- $\beta_2$ controls for prior-year level — isolates pure swing effect
- LP extension: $Y_{i,t+h}$ at $h = 0, 1, 2, 3$
- County + year FE; state-clustered SEs

# First Differences as a Design Choice: What Exactly Are We Estimating?

## The difference between level effects and swing effects

### Level effects (Phase 2)

- Asks: do counties that experience **more extreme shocks** in a given year have higher health costs that year (or 1–2 years later)?
- Identifies the effect of shock *occurrence* vs. no shock
- Cannot detect whether **rapid change** in climate exposure independently matters

### Swing effects (Phase 3)

- Asks: does a county that shifts sharply **from one year to the next** bear extra costs, *controlling for* where it was last year?
- Identifies the marginal effect of **volatility**
- Relevant for adaptation: a system that adapts to a persistent new climate level may still be damaged by rapid oscillation

### A climate science analogy

Compare two counties, both experiencing an "extreme heat" year (High CDD $= 1$). County A has had extreme heat for five consecutive years. County B is experiencing it for the first time after a run of mild summers. **Phase 2 gives both the same treatment indicator**. Phase 3 asks whether the *shock of the transition* carries its own financial cost — for B, $\Delta$CDD is large; for A, it is near zero.

- **Key insight from our data**: AQI level effects are consistently insignificant; AQI *swings* are significant at all four LP horizons
- This suggests health systems have partially adapted to local chronic pollution but are **not adapted to sudden AQI shifts**

# Weather Swings Affect Premiums, Debt, and Hospital Costs

**Primary ΔFE spec (h=0, unweighted);** $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$

## ΔHDD → Silver Premium

| Spec | Estimate | $p$ |
|---|---|---|
| ΔHDD ($h$=0) | +0.033*** | 0.0004 |
| HDD Onset (binary) | +49.46** | 0.0033 |
| HDD_Pos (asymm.) | +0.065* | 0.028 |

Cold

swing raises premiums on impact;
onset of first extreme-cold year
adds ∼ $49 to Silver benchmark.

## ΔZ_Temp → Medical Debt Median

| Horizon | Estimate ($) | $p$ |
|---|---|---|
| $h$=0 | +13.2* | 0.014 |
| $h$=2 | +19.4* | 0.033 |
| $h$=3 | +29.7*** | 0.0001 |

Warming swing

raises median debt
with a growing 3-year lag.

## ΔMax_AQI → Hospital Bad Debt (per capita)

| Horizon | Estimate ($) | $p$ |
|---|---|---|
| $h$=0 | +0.0050** | 0.0014 |
| $h$=1 | +0.0082*** | <0.001 |
| $h$=2 | +0.0080* | 0.015 |
| $h$=3 | +0.0157*** | <0.001 |

AQI swing effect

on bad debt **grows** across
all four horizons — a persistent ratchet.
Level effect was not significant in Phase 2.

## ΔZ_Precip / ΔPDSI → PCPI

| Exposure | $h$=1 est. | $p$ |
|---|---|---|
| ΔZ_Precip | −274*** | 0.0006 |
| ΔPDSI | −226*** | 0.0001 |

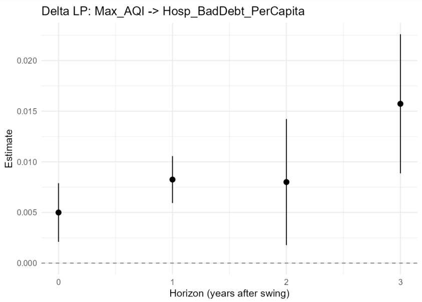
Precipitation and

drought swings
suppress per-capita income for 1–3 yrs.

# Swing Effects Grow Over Time: Two Key LP Profiles

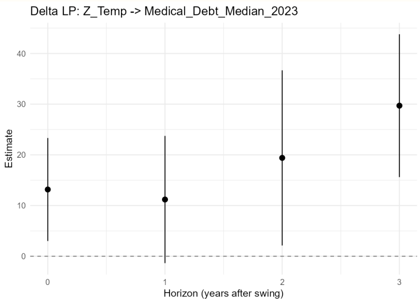
**Local projection estimates at $h = 0, 1, 2, 3$; county + year FE; state-clustered SEs; unweighted**

## $\Delta$Max_AQI $\rightarrow$ Hospital Bad Debt ($/capita)




Effect grows at every horizon — AQI bad-debt ratchet. Level effect was not significant in Phase 2.

## $\Delta$Z_Temp $\rightarrow$ Median Medical Debt ($2023)




Warming swing accumulates into medical debt; +$30 at $h$=3 ($p$<0.001).

# Asymmetry and Ratchets: Not All Swings Are Reversible

Asymmetric spec ($\Delta^+$ vs. $\Delta^-$) and onset/exit indicators

## Asymmetric Findings

- **AQI ratchet:** improving air quality still raises bad debt ($\Delta^-$, $+0.010^{***}$) — costs do not unwind when pollution falls
- **Drought worsening:** $\Delta^-$PDSI drives PCPI loss ($-207^{***}$); recovery asymmetrically weaker
- **CDD income:** symmetric — cooling-sector PCPI moves with swings in both directions ($\pm 3.3$, $p<0.001$)
- **Drought exit** *does* reduce medical debt ($-\$54^{*}$)

## What the Delta Analysis Adds

| Relationship | Finding |
|---|---|
| AQI → hosp. bad debt | **New:** swing sig.; level not |
| HDD → premiums | Confirms; adds onset pricing |
| Drought → income | Confirms; worsening-only |
| Temp swing → med. debt | New: 3-yr growing |
| Precip swing → PCPI | New: 3-yr suppression |

**Key insight:** weather volatility carries independent health-finance costs beyond level effects.

# Cross-Level Synthesis

What do state, county, and swing analyses agree on?

# State and County Pipelines Tell a Consistent Story

## Key findings replicated across both levels of analysis

| Finding | State | County |
|---|---|---|
| Cold / HDD → medical debt (prevalence & severity) | ✓ | ✓ |
| Cold / HDD → higher premiums | ✓ (via drought) | ✓ ($p<0.01$) |
| Cold / HDD → hospital bad debt ↑ | — | ✓ ($p<0.01$) |
| Drought → reduced Medicaid / PCPI | ✓ ($p<0.01$) | ✓ (PDSI) |
| Heat → household income loss | — | ✓ (Z_Temp***) |
| Heat lag → Medicare / premiums ↑ | ✓ (CDD***) | ✓ (CDD***) |
| Heat lag → hospital bad debt ↓ | — | ✓ ($p<0.001$) |
| Poor air quality → higher premiums | ✓ ($PM_{2.5}/O_3$***) | not modeled |

# Caveats and Limitations

## What we can and cannot claim from this design

### What two-way FE cannot rule out

- **Time-varying confounders**: a county-specific policy change (e.g. Medicaid expansion timing) that correlates with climate variation would contaminate the estimate. We control for observed time-varying covariates but cannot include everything.
- **Reverse causality**: unlikely here (climate is exogenous), but migration responses to climate shocks could affect measured outcomes.
- **Data quality**: hospital bad debt is self-reported; $\sim$23% of county-years are missing. Medical debt from credit bureaus is a snapshot, not a flow.

### Scope of claims

- Results are **reduced-form associations** under two-way FE identification — not causal in the IV sense
- **Short-run effects** only: the panel is 2011–2023. Long-run adaptation responses (e.g. insurer exit, hospital closure) are not captured
- **Weather swing analysis** captures short-run responsiveness to volatility; coefficients do not generalise to slow climate trends
- The medical debt reporting-rule exclusion (CO 2023) is conservative; it reduces rather than inflates the debt findings

**Bottom line:** results are robust across multiple specifications, weighting approaches, and levels of analysis — but should be interpreted as evidence of association, not a controlled experiment.

# Policy Implications

What should adaptation planners, insurers, and health systems take from this?

# Policy Implications: Climate Adaptation Must Include Health Finance

Three actionable implications for adaptation policy, insurer regulation, and health system design

## Adaptation policy

**Current gap:** most adaptation frameworks target mortality and infrastructure.

**Finding:** cold, drought, and AQI shocks increase medical debt and reduce income for 1–3 years after the event.

**Implication:** **climate resilience funds** should include a health-finance window — covering the period when households are financially recovering from a shock, not only emergency response.

## Insurance regulation

**Finding:** insurers raise ACA premiums in extreme cold years (+\$28–\$86) and after extreme heat (+\$21–\$35 at 1–2 yr lag). The swing analysis shows a cold *onset* adds ∼\$49 instantly.

**Implication:** regulators should **monitor climate-driven premium volatility** as a component of rate review. Counties transitioning into new climate regimes face a double burden: worsening shocks and rising insurance costs.

## Hospital systems

**Finding:** hospital bad debt rises +\$4–\$5/capita in cold years and continues to grow after AQI swings for three years.

**Finding:** drought worsening triggers income losses and employment falls simultaneously.

**Implication:** **climate stress-testing** for hospital finances should use local climate volatility indices, not just long-run averages. Systems in high-volatility corridors need expanded charity-care reserves.

# Next Steps

Planned extensions to the empirical framework

## Empirical Extensions

- **Heterogeneity:** rural vs. urban and high- vs. low-AQI subsamples for both level and swing analyses
- **Mechanisms:** separate respiratory-illness from energy-poverty channels for cold shocks
- **County unemployment:** replace the ACS proxy with BLS LAUS series
- **Policy counterfactuals:** ACA Medicaid expansion $\times$ climate shocks

## Writing and Defence

- **Chapter draft:** unify state, county, and swing narratives into a single empirical chapter
- **Swing robustness:** population-weighted LP profiles for delta results; cumulative IRF plots
- **Thesis write-up:** convert synthesis tables into publication-ready regression tables